Chegofatjo-ya-Modimo Mawela

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EDUCATION

University of Cape Town Cape Town, South Africa
BSc Computer Science and Applied Statistics January 2024 – Present
Current GPA: 84.20

SKILLS SUMMARY

Languages: Python, Java, SQL, R, HTML, CSS, JavaScript
Frameworks: Pandas, NumPy, Scikit-learn, PyTorch, FastApi, React, React Native
Tools: Excel, MySQL, SQLite, LaTeX
Platforms: Jupyter, VS Code, R Studio
Soft Skills: Communication, Teamwork, Problem-solving

WORK EXPERIENCE

CS1 Hotseat Tutor January 2026 – Present
Department of Computer Science — University of Cape Town
- Provide academic support to all first-year Computer Science students
- Assist students with conceptual understanding, algorithm design and coursework guidance

STA1000F/S Tutor February 2025 – Present
Department of Statistical Science — University of Cape Town
- Assist students in understanding fundamental concepts in statistics
- Support student improvement through tutorials, whiteboards and hot seats
- Collaborated with tutoring team to re-structure course layout and structure for optimal learning (2025)

CSC1015F & CSC1016S Tutor February 2025 – December 2025
Department of Computer Science — University of Cape Town
- Facilitated weekly practical sessions in Python and Java programming
- Marked programming assignments and assessments

E-commerce store manager March 2020 – December 2020
The Masked Magician — Etsy
- Built and scaled a dropshipping e-commerce store
- Generated nearly R65000 in 5 months

PROJECTS

Redman 3.0 March 2025 – Present
Github Link (results)
- Building a data-driven expert advisor for financial market instrument forecasting
- Uses human psychology (via social media scraping), news sentiment, country personas, statistical methods (Bayesian reasoning, ARIMA modeling etc.) and machine learning
- Currently in development, early version results are linked above

World Development Indicator Analyser July 2025 – August 2025
GitHub Link
- Built a K-means based analyser which clusters countries into multiple personas using the World Development Indicators
- Implemented a custom Pearson correlation coefficient based initialization method
- Used a custom "running average" imputation strategy, which improved clustering performance
- Classifies countries into personas for use in investment portfolio construction alongside correlation analysis

Messaging app February 2026 – March 2026
GitHub Link
- Developed a client-server messaging application using Python socket programming
- Designed and implemented a custom protocol for reliable message and file transfer
- Built client-side functionality including local message storage using SQLite for improved performance
- Supports one-on-one and group chat rooms with integrated file sharing